Daily Dev Mix

Group Members (GitHub):

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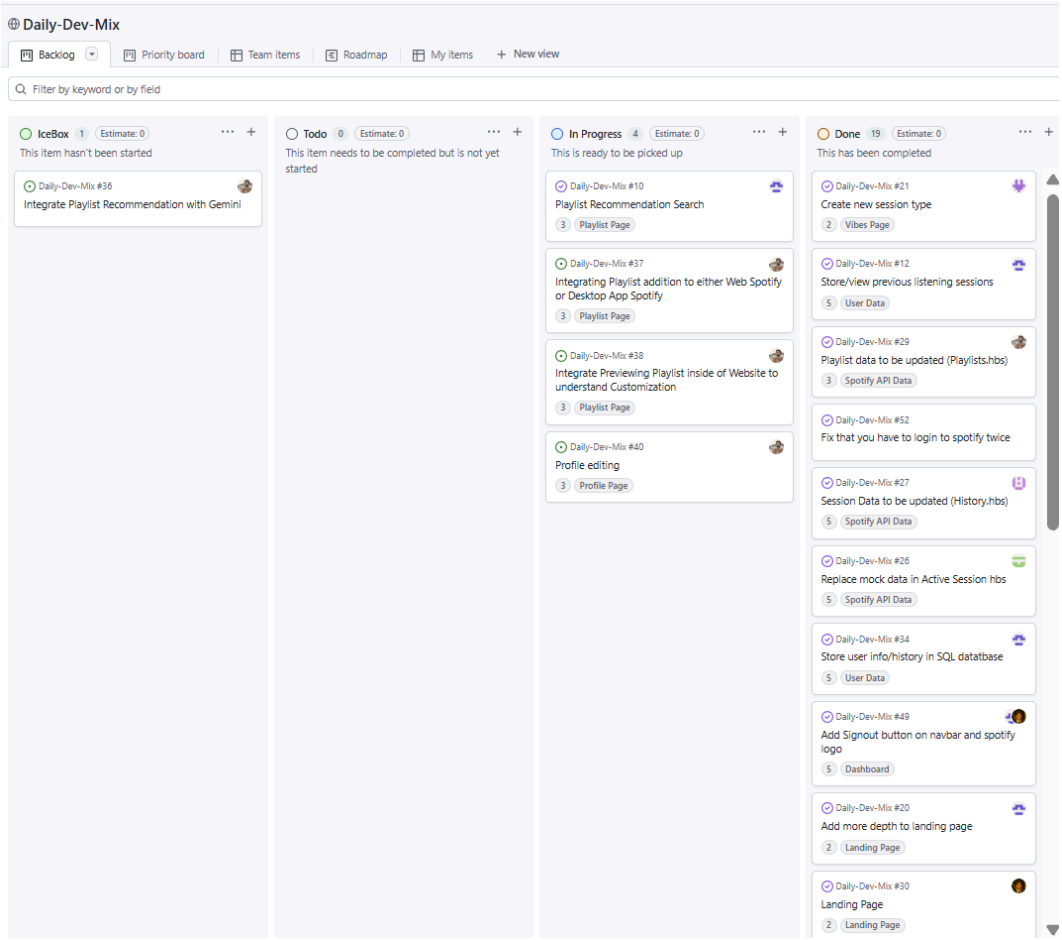
Project Description:

Daily Dev Mix is a Spotify companion tool whose main goal is to create personalized playlists for the user based on activities they do while listening to music. The user signs into the website using their Spotify account before being redirected to the dashboard. The user is then able to select a predefined or user-created “vibe” or activity they will be doing as they start their listening session. Once the “vibe” is selected and the start session button is clicked, the user can start listening to music on Spotify, and their listening data will be displayed and collected on the site using Spotify API calls. After the session, a playlist is created based on the music that was listened to during that activity. These playlists are built on with each successive listening session and are separated based on the different activities. Spotify does have special activity playlists built in, but our API aims to create personalized playlists based on the music listened to during the session, so users can enjoy music they actually want to hear while engaging in their activity.

Project Tracker:



https://github.com/orgs/Daily-Dev-Mix/projects/4

Video:

https://drive.google.com/file/d/1ULiOOnj9ALZfUmMMMKN5xbVdeymbtB7u/view?usp=sharing

VCS:

https://github.com/Daily-Dev-Mix/Daily-Dev-Mix

UI/UX Design Summary

This section summarizes how the Daily Dev Mix meets the project's creative and functional design standards.

1) Visual Design & Cohesion
    a) The site features a professional dark mode aesthetic. We maintained a strict palette of black and the Spotify signature green, paired with consistent border radii and typography across all sections to ensure a unified brand identity.
2) Navigation & Flow
    a) Navigation is easy and intuitive, guiding users between pages and tasks seamlessly, ensuring the core task is completed without friction.
3) Custom Styling
    a) We utilized custom grid architectures. Notable styling includes card effects, button gradients, and a responsive layout that adapts seamlessly to any screen size.
4) Engagement
    a) The interface incorporates dynamic feedback elements, such as hover transitions and custom iconography.
5) Accessibility
    a) The site was built with readability and inclusivity in mind. All colors were carefully picked to contrast, and keep a constant theme that would still look good.

Innovative Problem Selection Summary

1) Target Audience: Students, developers, and professionals who use Spotify as a productive tool who are tired of Spotify's daily mixes that are repetitive and fail to adapt and recommend the correct music for the task the user is currently doing.
2) Evidence of the Problem: While Spotify offers millions of songs, users often feel overwhelmed by generic playlists.

    Validation: We conducted brief interviews with a small group of peers. Most of the participants noted that Spotify's playlists such as "Focus" often include tracks that lack a personal connection to the user's taste and feel too generic while repeating a lot of their listening history instead of focusing on the mood.

    Clearly, there exists a demand for a tool that filters a user's own history in specific situations, rather than repeating random music from their listening history, or in some cases, not even from their listening history, but completely new, unfamiliar music.

3) Insufficiency of Existing Solutions

    Current solutions like Spotify's "Daylist" or "Daily Mix" tell the user what they should like based on the time of day, which is too predictive and not personal enough.

    Limitation: If a user is a fan of Heavy Metal but likes to study to Ambient music, Spotify's "Study" recommendation will either be generic Ambient, which is not personal or Metal, which is not what they study to typically.

    Our Solution: By allowing the user to select the "Vibe" first and then pulling metadata from their actual history, Daily Dev Mix ensures the music matches the task and also resonates with the user's personal preferences.

4) Differentiation From Simple “Tutorial-Style” Apps

Unlike a basic tutorial that simply displays top tracks, this application:

i. Consumes live listening data via the Spotify API.

ii. Maps user artists and genres to a specific mood or user-selected “vibe”

iii. Systemically creates and names a new playlist based on the filtered data

5) Intentional Testing and Observations

We conducted tests to our target audience across 5 use cases, leading to critical iterations:

i. Bug Fixes: Identified and fixed an OAuth configuration error and a bug where "highlighted" vibes weren't being selected by the program.

ii. UX Improvements: Based on user confusion, we included an explanation guide and error handling to ensure metadata retrieval through users’ understanding of how to play music in Spotify.

iii. Validation: Ensured database integrity by verifying that our "Vibe" creator successfully blocked invalid inputs such as missing emojis.

Contributions:

Ian Martin: During the project, I first worked on setting up the handlebars backbone structure for the website, including the necessary partials, layouts, and pages that were later built on by the team with the UI components. Additionally, I set up parts of the index.js to render and redirect pages, as well as some basic Spotify API calls that were used to implement the listening tracking

features. For the UI, I created the responsive nav bar with an integrated logout button. I also created the databases that were used to store the user and listening data.

Devang Pandey: My work began by setting up the initial and organizational repositories. I imported the initial Figma UI, which established the app's frontend and backend structure. A major portion of my contributions focused on integrating the Spotify API, implementing features such as saving tracks, adding songs to playlists, syncing user listening data, and enabling playlist export within the app. I also developed profile and playlist management tools, refined user-facing settings to be more intuitive, and made ongoing UI improvements across core pages.

Gregory Lowman: At the beginning of our project, we had a strong idea for what we wanted the final product to look like. We had an interactive prototype that had all of the styling options we were interested in, along with some minor functionality. I recreated the prototype using Handlebars and HTML so that we could have a shell to begin to work in. It used mock data for the group to populate as we go along with the pages linking and the Spotify API being fully linked. Once the app was put together (for testing), I launched the app on Railway by linking it to the main GitHub branch. In doing this, we have been able to view changes in real time and consistently make sure everything is functioning properly.

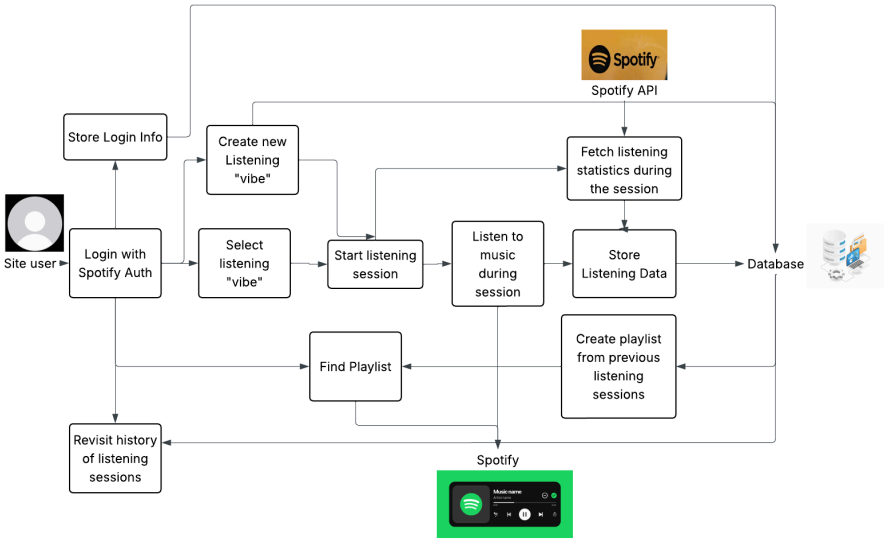
Matthew Aldridge: My work at the start of the project was mainly focused on helping others get the app running and giving us something to look at. Most of my work time was focused on creating or modifying the UI. I worked on giving the user the ability to create and delete different types of listening sessions. I made a modal that allows you to pick a name and add an emoji to

your listening session type. I also added a trash button to each session type once it was created, which allows you to get rid of session types.

Vera Zaric: In the beginning of the project, I focused on coordinating with teammates about what needed additional development. I focused a lot of my time on working on features for the UI. For example, I created a feature for the homepage that welcomes the user with a different random phrase every time the page is reloaded.

Dylan Long: I worked primarily as a frontend developer. My main contribution was the final landing page, which I made from scratch. I also helped debug routing in our server to make sure everything connected seamlessly with our frontend. Furthermore, I polished the dashboard/home pages of the app, adding aesthetically pleasing UI elements such as confetti when playlists were made, or polishing the dashboard navbar/ dashboard overall. Most of my work was in the handlebar files.

Use Case Diagram:



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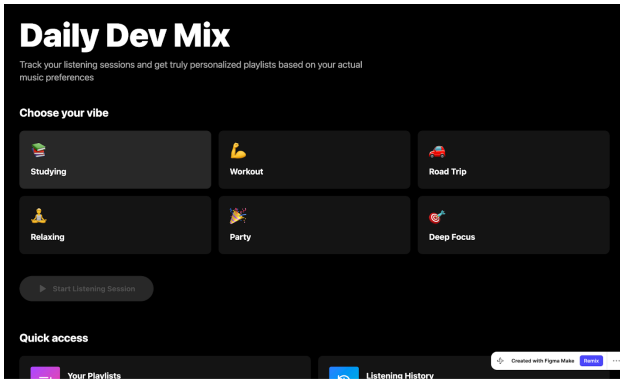

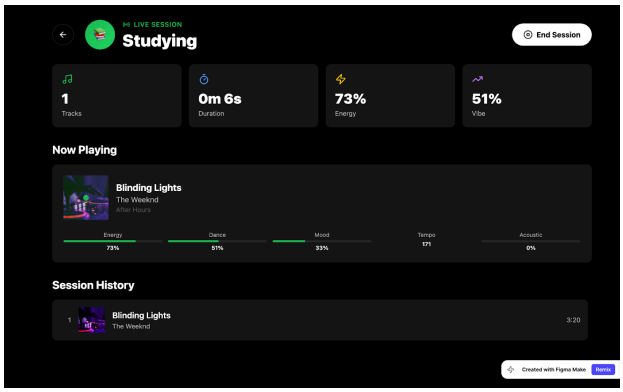

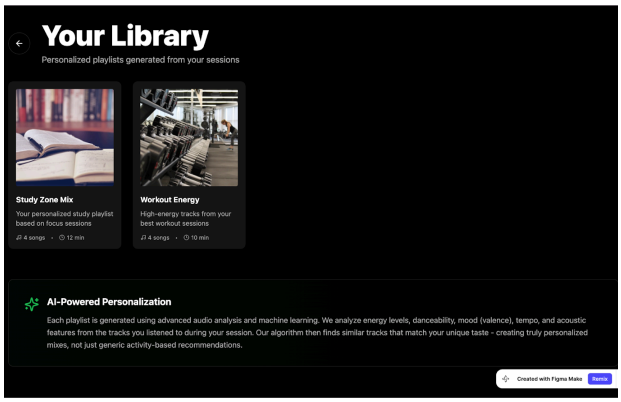

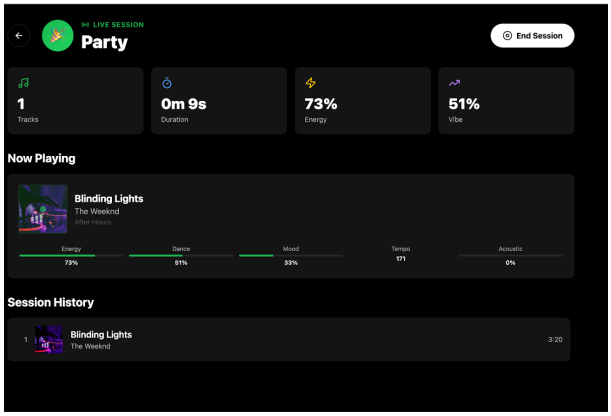

User Acceptance Test Results:

Use Case 1: User Spotify Login

- What Users did:
  The users were all able to navigate to the login button and successfully connect their Spotify account, signing them in to the website. One user did accidentally hit back instead of connect on the Spotify pop-up, which just made them have to redo the process with no errors. When logging in one user also had to select connect twice as the first time did not fully log them in.
- Users' Reasoning:
  The users explained that the sign-in process was pretty intuitive and easy to complete. One user said the process was very similar to when they sign in to sites using Gmail, which made it easy and successful. The one user who accidentally hit the back button explained they were just trying to go too quickly through the process.
- Behavior Consistency with Use Case:
  The results of the tests fit our initial expectations for the use case, even with our consideration of the negative case of the user canceling the sign-in. The only main difference was that the user who had to complete the process twice to successfully sign in.
- Reason for Deviation:
  The reason for that difference was an error that the team needed to work through to correct.
- Changes Made:
  The team found the issue with the OAuth configuration to fix the error, requiring users to confirm their Spotify account twice in order to log in.

Use Case 2: Tracking a Listening Session

- What Users did:
  The users generally knew exactly what to do and selected one of the available "vibes" and then started the session. One user did not actually click a "vibe" before starting a session and was confused why it didn't work. After they started their listening sessions they played songs on Spotify, skipping around to see the app recognize the change the song and then were all able to successfully end the session which redirected them to the playlist they created.

- Users' Reasoning:

They reported that the process was pretty simple to complete but once they start a session they said it wasn't totally obvious that they had to play music on the Spotify app instead of just our site if they weren't aware of how it worked before hand. The user that tried to start the session without clicking a "vibe" said it looked like that "vibe" was already selected, so they thought they could just start with that.

- Behavior Consistency with Use Case:
  The overall process worked as intended, as all the users were able to start a session after selecting a "vibe" and the app successfully tracked the music they listened to on Spotify during this period and it was displayed on the dashboard. We did not expect the issue of the session not starting with the preselected "vibe".
- Reason for Deviation:

  The reason for the deviation with the auto-selected "vibe" was that we thought that a user would just always click a "vibe" before starting each time. We also thought that if they didn't select one, the app would default to selecting the previously used one, but we discovered a bug that it does highlight this "vibe," but it is not actually selected until the user clicks.
- Changes Made:
  To adjust to the results from testing we have updated the landing page to fully explain the process of the app. We have also been working through the bug of the "preselected vibe." to have it actually be selected, not just highlighted. In addition, we plan to add an error Message for when a "vibe" is not selected, and a message explaining to the user that they need to start listening to music on the Spotify app when the session has started.

Use Case 3: Playlist Generation from Listening Session

- What Users did:
  Continuing from the test above, users were able to end their listening session, which redirected them to the playlist page with a new playlist created. The users then clicked into the playlist that was just generated, and that redirected them to the Spotify page where their full generated playlist was displayed. The users examined the playlists and generally said that the playlists only contained songs from the artists played during the session, and they would prefer if more similar artists were added as well.
- Users' Reasoning:
  The users liked how the website took them right to the playlists when the listening session ended, making it easy to navigate to the playlist. They also noted that clicking the playlist made sense, and they liked how it took them to Spotify, though it wasn't necessarily expected by the user.
- Behavior Consistency with Use Case:
  For this test, the results lined up with how we intended the website to work. The site successfully created a playlist and redirected them to Spotify when clicked.

- Reason for Deviation:
  There wasn't any deviation.
- Changes Made:
  No changes were made to this feature, though the team is trying to work on a solution to have a wider range of similar songs suggested to the user in the created playlist, but has run into issues with deprecated tools on the Spotify API.

Use Case 4: Create custom "vibe"

- What Users did:
  The users all selected the "add topic" button, which opened the modal in which they typed the name of a "vibe" they wanted to add, as well as clicking an emoji to accompany the topic. One user did attempt to save the topic before selecting an emoji, but the website blocked the modal submission and required the user to go back and select an emoji. After they all submitted the modal the topic was successfully added to the topic list for the users to select in an upcoming listening session.
- Users' Reasoning:
  Most users found this process pretty self-explanatory and easy to use. The user who tried to submit the modal without an emoji explained they did not find an emoji that fit the activity, so they thought they could add it without one at all.
- Behavior Consistency with Use Case:
  These results lined up to what we expected. The website successfully added the topics that were created and passed the negative test of invalid input.
- Reason for Deviation:
  There was no deviation.
- Changes Made:
  No changes were made to the application.

Use Case 5: Logout of Spotify Account

- What Users did:
  All users simply hit the log-out button and were successfully logged out and taken back to the landing page.
- Users' Reasoning:
  The users reported that the logout button was placed in a clear and accessible spot which made the process easy.
- Behavior Consistency with Use Case:
  This test lined up with our intended results as all users could click the logout button and were signed out of their account and redirected to the landing page quickly.
- Reason for Deviation:
  There was no deviation.
- Changes Made:
  No further changes were made since the feature executed correctly.

Deployment:
https://daily-dev-mix-production.up.railway.app